

# Undergraduate Research in AI/ML for Aerospace Systems

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## Overview

Using Andrej Karpathy's "Zero to Hero" playlist, develop jupyter notebooks covering the contents of the video. Use the videos to design a small action language model (SAM) to optimize a simulation based on extraction and pick ups modelling several AOE vehicles for time optimization.

## Preliminary Schedule

| Week          | Research Schedule                                                                           | Important Dates                                       |
|---------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------|
| FA01 (Aug 19) | Set up research credit<br>4994 Syllabus                                                     |                                                       |
| FA02 (Aug 26) | Set up Linux connection and notebook<br>Connect Starbelt WSL remote host                    |                                                       |
| FA03 (Sep 02) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |
| FA04 (Sep 09) | Continue jupiternotebooks of zero to hero<br>proofread 01, then develop 02 code and writing |                                                       |
| FA05 (Sep 16) | Continue jupiternotebooks of zero to hero<br>develop 03 04                                  |                                                       |
| FA06 (Sep 23) | Continue jupiter notebooks of zero to hero<br>develop 05 06 07 08                           |                                                       |
| FA07 (Oct 07) | Continue jupiter notebooks of zero to hero<br>develop 09 10 11(makemore done)               |                                                       |
| FA08 (Oct 14) | Continue jupiter notebooks of zero to hero<br>Incorporation with Julia                      |                                                       |
| FA09 (Oct 18) | Continue jupiter notebooks of zero to hero                                                  |                                                       |
| FA10 (Oct 25) | End milestone 2<br>Slides, written summary                                                  |                                                       |
| FA11 (Nov 01) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |
| FA12 (Nov 08) | A short description of the weekly task<br>The deliverable(s) for the week                   | DD Mon: Conference name<br>Description of what is due |
| FA13 (Nov 15) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |
| FATB (Nov 22) | A short description of the weekly task<br>Thanksgiving break                                |                                                       |
| FA14 (Nov 29) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |
| FA15 (Dec 06) | End milestone 3<br>Slides, written summary                                                  |                                                       |
| FA16 (Dec 13) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |
| GR17 (Dec 20) | A short description of the weekly task<br>The deliverable(s) for the week                   |                                                       |